

Finite element analysis of punching shear of concrete slabs using damaged plasticity model in ABAQUS



Aikaterini S. Genikomsou*, Maria Anna Polak

Department of Civil and Environmental Engineering, University of Waterloo, Waterloo, ON, N2L 3G1, Canada

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ABSTRACT

Nonlinear finite element analyses of reinforced concrete slab-column connections under static and pseudo-dynamic loadings were conducted to investigate their failure modes in terms of ultimate load and cracking patterns. The 3D finite element analyses (FEA) were performed with the appropriate modeling of element size and mesh, and the constitutive modeling of concrete. The material parameters of the damaged plasticity model in ABAQUS were calibrated based on the test results of an interior slab-column connection. The predictive capability of the calibrated model was demonstrated by simulating different slab-column connections without shear reinforcement. Interior slab-column specimens under static loading, interior specimens under static and reversed cyclic loadings, and edge specimens under static and horizontal loadings were examined. The comparison between experimental and numerical results indicates that the calibrated model properly predicts the punching shear response of the slabs.

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1. Introduction

Punching shear failure is caused by high shear stresses in the slab-column connection area of reinforced concrete flat slabs. This brittle failure was examined by many researchers in the form of tests, analytical models, and finite element analyses. Several researchers proposed empirical equations based on tests observations [1–4], which provide the basis of the existing design codes [5,6]. A brief review of punching shear in slabs without shear reinforcement begins with Elstner and Hognestad [1] and Moe [2] who performed experimental work that led to the ACI design provisions [5]. In Europe, Regan [3] and Regan and Braestrup [4] proposed empirical equations, that are the basis for the current European design approach (EC2) [6] for punching shear. The existing punching shear testing database, even though it is large [1–4], cannot address all aspects of punching shear stress transfer mechanisms. Therefore, in modern research in structural engineering, finite element analyses (FEA) are essential for supplementing experimental research in providing insights into structural behavior, and, in the case presented herein, on punching shear transfer mechanisms. Nonlinear FEA can show crack formation and propagation, deflections, possible failure mechanisms and supplement experimental observations, where the test measurements are not known. However, the complexity of the nonlinear finite element models

is inherent due to various theories used in material modeling, element selection and solution procedures that these models include. Many different constitutive models have been utilized in finite element simulations, among others, the most known are: nonlinear elasticity, plasticity, damage mechanics and coupled damage and plasticity models [7–13]. Research on layered shell finite element analysis of punching shear was performed by Polak [14,15] and Guan and Polak [16].

The work described herein, is on modeling concrete slab-column connections using a 3D analysis with the commercial FEA program ABAQUS. The coupled damaged-plasticity model for 3D finite element analysis, which is offered in ABAQUS [17], was adopted for the representation of concrete. The concrete damaged plasticity model is coupled with the fictitious crack model introduced by Hillerborg [18]. This is an energy criterion based on the fracture energy that should prevent mesh-sensitivity and allow for numerical convergence. The adopted finite element model was calibrated based on the selected experimental results.

Five slab-column specimens (SB1, SW1, SW5, XXX and HXXX) without shear reinforcement were analyzed. The slab, SB1, is an interior slab-column connection that was tested under static loading through the column [19]. The slabs, SW1 and SW5, are interior slab column connections that were tested under gravity static loading through the column and pseudo seismic horizontal loading [20]. Finally, the specimens, XXX and HXXX, are edge slab-column connections that were tested under vertical loading through the column and an unbalanced moment at the columns [21].

* Corresponding author.

E-mail addresses: agenikom@uwaterloo.ca (A.S. Genikomsou), polak@uwaterloo.ca (M.A. Polak).

The calibration of the model, based on tested slab SB1, control specimen, [19] is presented first. The sensitivity of the material and the FEA model to various parameters is discussed. The constitutive model is described in detail, including the effects of various material parameters on the accuracy of the analysis. Then, the finite element simulation results are presented for the reinforced concrete slab-column connections under various load combinations [20,21]. The numerical results are compared to the test results in terms of deflections, strength and crack patterns. The aim of this paper is to present the effectiveness of the proposed calibrated finite element model in describing and analyzing punching shear tests by identifying key parameters of the model.

2. Test specimens

The test specimens used for the finite element analyses had no shear reinforcement and the height of all slab specimens was 120 mm. These, were isolated slab-column connections, loaded through the column and simply supported along the edges that represented the lines of contra flexure in the parent slab-column system. The first analyzed specimen is the interior connection (SB1) that was tested under static loading through the column. The height of the column extending from the top and the bottom faces of the slab was 150 mm. Then, the two interior slab-column connections (SW1, SW5) that were tested under gravity static and pseudo seismic loadings were analyzed. These slabs were loaded in two stages. In the first stage, a vertical load was applied through the top column with a loading rate of 20 kN/min. The slab SW1 was loaded up to 110 kN vertical load, the slab SW5 was loaded up to 160 kN. Then, the vertical loads were kept constant and the two horizontal actuators started to apply horizontal drift to the top and bottom columns at a distance 565 mm from the slab's faces following a loading path described in [20]. The total height of the columns was 700 mm. The gravity shear ratio, V/V_n , for the slab SW1 was 0.54 and for the slab SW5 0.68, where

$V_n = 0.33 \cdot \sqrt{f'_c} \cdot b_o \cdot d$ (MPa), b_o denotes the perimeter length of the critical section and d the effective thickness of the slab equal to 90 mm. All interior connections had overall dimensions in plan 1800×1800 mm with simple supports at 1500×1500 mm. Corners were restrained from lifting. Finally, the two edge slab-column connections (XXX, HXXX) were analyzed. These slabs were tested under a vertical shear force (V) that was applied on the top of the upper column and two horizontal forces (H), leading to the unbalanced moment, that were applied to the columns in three stages at a distance 600 mm from the slab's faces. The total height of the columns was 700 mm. The slabs' in-plane dimensions were 1540×1020 mm. In the first stage of testing, the loads were increased with a rate of 2.5 kN/min. until reaching the service load, $V = 43$ kN. Then the loads were cycled 10 times between the dead loads and the dead plus the live loads, in order to simulate the repetition of the live loads. At the final stage, the loads were increased at 1.5 kN/min. rate until failure. The ratio between the unbalanced moment (M) produced by the two horizontal forces (H) and the vertical shear force (V) was equal to 0.3 m for the specimen XXX and 0.66 m for the specimen HXXX. These ratios were kept constant during the whole loading process. The dimensions of the specimens and the loading process are presented in Fig. 1. The reinforcement configuration of each specimen can be found in [19–21].

The material properties of each tested slab are presented in Table 1. The compressive strength of concrete was found from the concrete cylinders, tested at the time of the slabs' tests (over 28 days), and the tensile strength was obtained from the split cylinder tests. The yield strength for the tension and compression longitudinal reinforcement was the same for the slabs SB1, SW1

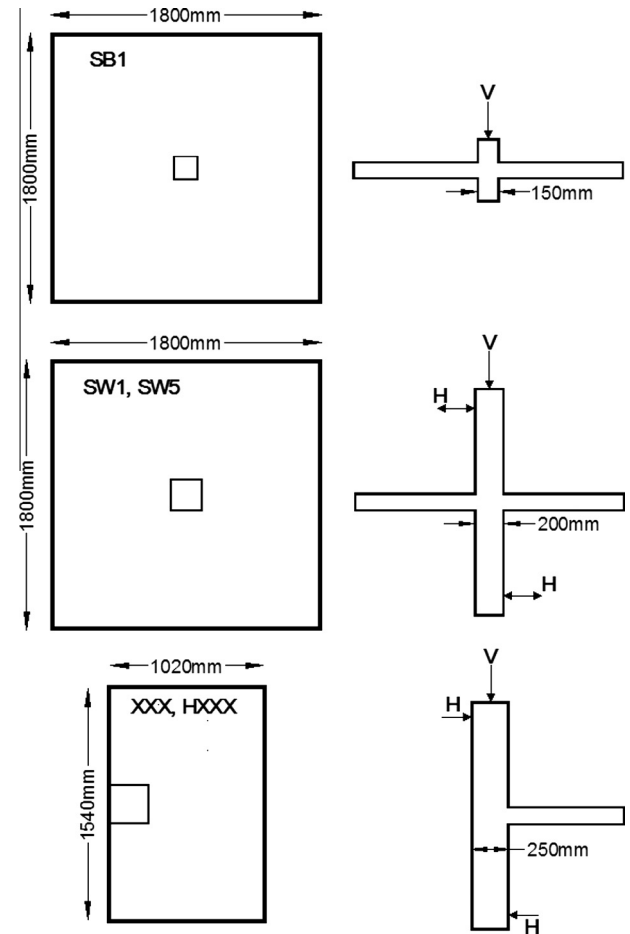


Fig. 1. Schematic drawings of the specimens – dimensions and loading.

Table 1

Material properties of the tested slabs.

Slab specimen	Compressive strength of concrete (MPa)	Tensile strength of concrete (MPa)	Yield strength of flexural reinforcement (MPa)
SB1	44	2.2	455
SW1	35	2	470
SW5	46	2.2	470
XXX	33	1.9	545 (tension), 430 (compression)
HXXX	36.5	2	545 (tension), 430 (compression)

and SW5. Slabs XXX and HXXX had different yield longitudinal strength for the tension and compression reinforcements.

All specimens failed in punching shear. The information regarding their failure loads and comparisons with the simulation results, are presented in the following sections.

3. Finite element simulations

3.1. Methodology

By considering specimens' symmetry, one quarter of the control specimen SB1 and half of all the other slabs (SW1, SW5, XXX and HXXX) were used for the simulations. 8-noded hexahedral (brick) elements were used for concrete with reduced integration

(C3D8R) to avoid the shear locking effect [17]. 2-noded linear truss elements (T3D2) were used to model reinforcements. The embedded method was adopted to simulate the bond between the concrete and the reinforcement, assuming perfect bond. 6 brick elements were used through the thickness of the 120 mm slab's with all concrete elements having the same size of 20 mm. The specimen SB1 had 9211 mesh elements and 11,194 nodes, the specimens SW1 and SW5 had 22,028 mesh elements and 26,767 nodes and the specimens XXX and HXXX were meshed with 18,150 elements and 22,123 nodes. Restraints were introduced at the bottom edges of the specimens in the direction of the applied load. The summation of the reactions at the edges, where the boundary conditions were introduced, yielded the reactions equal to the punching shear loads. Fig. 2 gives details regarding the geometry and the boundary conditions of the specimens that were used for the simulations. The control specimen SB1 was analyzed using both static analysis in ABAQUS/Standard and quasi-static analysis in ABAQUS/Explicit. In the static analysis, a displacement was applied through the column stub. In the quasi-static analysis, a low velocity was applied. This, last type of analysis, was used for all specimens. The velocity was increasing with a smooth amplitude curve from 0 (mm/s) to a different velocity (mm/s) depending on the specific slab. Slabs SB1, SW1, SW5 and XXX were loaded by applying a velocity that increased from 0 mm/s to 40 mm/s, such that the slab displaced at a rate of 20 mm/s. Slab HXXX was loaded by applying a velocity that increased from 0 mm/s to 20 mm/s so as the center of the slab displaced at 10 mm/s. Among the constitutive models for simulating the behavior of concrete, the concrete damaged plasticity model that ABAQUS offers was chosen and a detailed description of this model is presented in the next section.

3.2. Concrete damaged plasticity model in ABAQUS

A brief presentation of the damaged plasticity model from ABAQUS is presented. The yield function was proposed by Lubliner et al. [22] and then modified by Lee and Fenves [23]. It is defined according to Eq. (1):

$$F = \frac{1}{1-\alpha} (\bar{q} - 3\alpha\bar{p} + \beta(\bar{\epsilon}^{pl})\langle\bar{\sigma}_{max}\rangle - \gamma\langle-\bar{\sigma}_{max}\rangle) - \bar{\sigma}_c(\bar{\epsilon}_c^{pl}) \quad (1)$$

Parameter α is calculated according to Eq. (2), where (σ_{b0}) is the biaxial compressive strength and (σ_{c0}) is the uniaxial compressive strength. The default value of the ratio $(\sigma_{b0}/\sigma_{c0})$ is 1.16, according to [17].

$$\alpha = \frac{(\sigma_{b0}/\sigma_{c0}) - 1}{2(\sigma_{b0}/\sigma_{c0}) - 1} \quad (2)$$

In Eq. (1), $\bar{p}$ is the hydrostatic pressure stress and $\bar{q}$ is the Mises equivalent effective stress. Function $\beta(\bar{\epsilon}^{pl})$ shows up in the yield function, when the algebraically maximum principal effective stress $(\bar{\sigma}_{max})$ is positive (the Macauley bracket $\langle \cdot \rangle$ is obtained as: $\langle x \rangle = \frac{1}{2}(|x| + x)$) and it is determined as:

$$\beta(\bar{\epsilon}^{pl}) = \frac{\bar{\sigma}_c(\bar{\epsilon}_c^{pl})}{\bar{\sigma}_t(\bar{\epsilon}_t^{pl})} (1 - \alpha) - (1 + \alpha) \quad (3)$$

where $\bar{\sigma}_c(\bar{\epsilon}_c^{pl})$ and $\bar{\sigma}_t(\bar{\epsilon}_t^{pl})$ are the effective cohesion stresses for compression and tension, respectively. In biaxial compression, where $\bar{\sigma}_{max} = 0$, the parameter $\beta(\bar{\epsilon}^{pl})$ is not active and the only parameter being is the parameter α .

The shape of the yield surface is defined by the parameter γ according to Eq. (4). Parameter γ is active in Eq. (1), when the

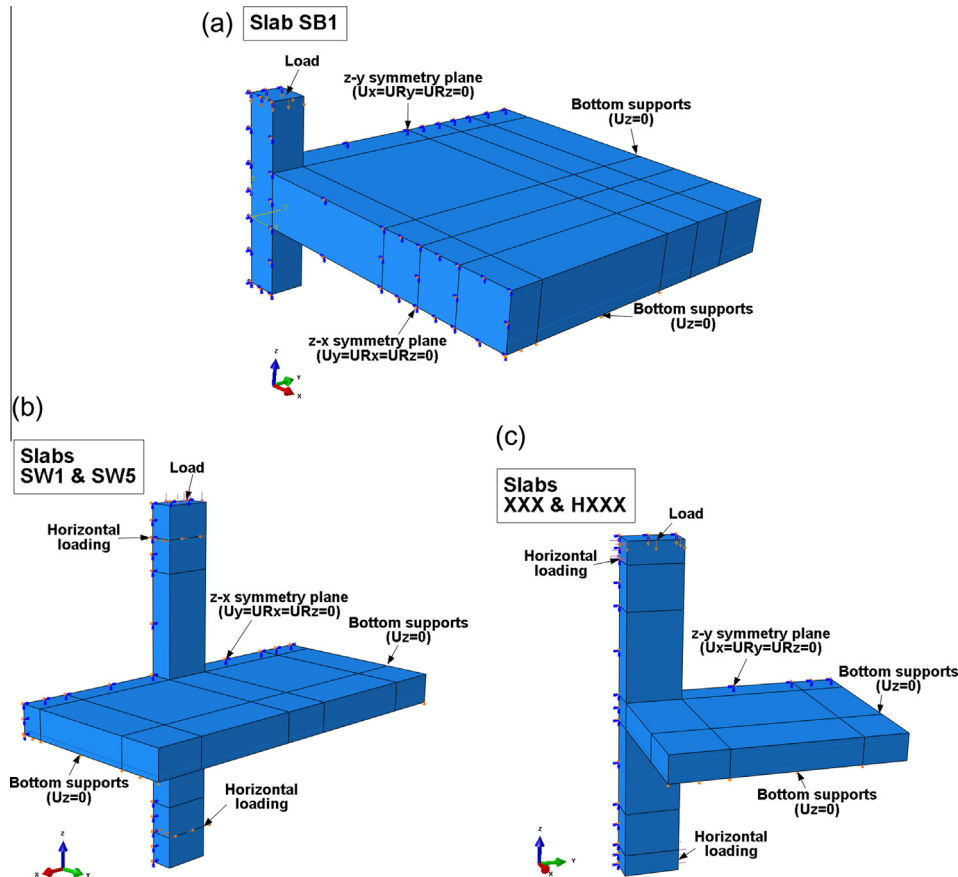


Fig. 2. Geometry and boundary conditions: a) SB1 slab; b) SW1 and SW5 slabs; c) XXX and HXXX slabs.

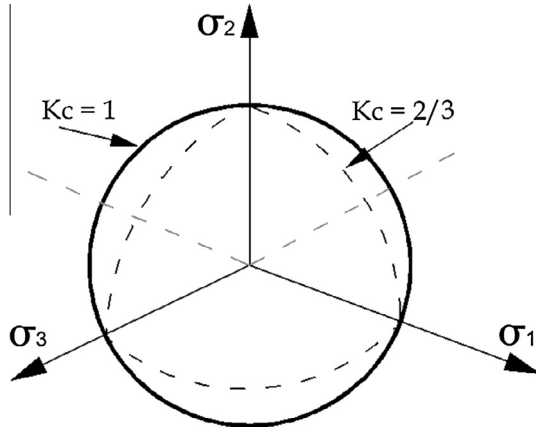


Fig. 3. Yield surfaces in the deviatoric plane ($K_c = 2/3$ corresponds to the Rankine formulation and $K_c = 1$ corresponds to the Drucker–Prager criterion).

maximum effective principal stress ($\bar{\sigma}_{max}$) is negative, happens in triaxial compression.

$$\gamma = \frac{3(1 - K_c)}{2K_c - 1} \quad (4)$$

K_c is the ratio of the tensile to the compressive meridian and defines the shape of the yield surface in the deviatoric plane (Fig. 3).

Concrete damaged plasticity model uses the flow potential function, $G(\sigma)$, which is a non-associated Drucker–Prager hyperbolic function and is defined according to Eq. (5).

$$G(\sigma) = \sqrt{(\varepsilon \sigma_{t0} \tan \psi)^2 + \bar{q}^2} - \bar{p} \tan \psi \quad (5)$$

In Eq. (5), ε is the eccentricity that gives the rate at which the plastic potential function approximates the asymptote, σ_{t0} is the uniaxial tensile stress and ψ is the dilation angle measured in the $p - q$ plane at high confining pressure. Fig. 4 shows the plastic potential function compared to the yield surface. The plastic strain increment is normal to the plastic potential function. In Fig. 5 is presented schematic the dilation angle and the eccentricity. According to [17] the default value for the eccentricity is equal to 0.1, shows that the concrete has the same dilation angle through a wide range of confining pressure stresses. The dilation angle shows the direction of the plastic strain increment vector. The non-associated flow rule means that the plastic strain vector is normal to the plastic potential function that differs from the yield surface.

Damage is introduced in the model according to Eq. (6).

$$\sigma = (1 - d)\bar{\sigma} = (1 - d)E_0 : (\varepsilon - \varepsilon^p) \quad (6)$$

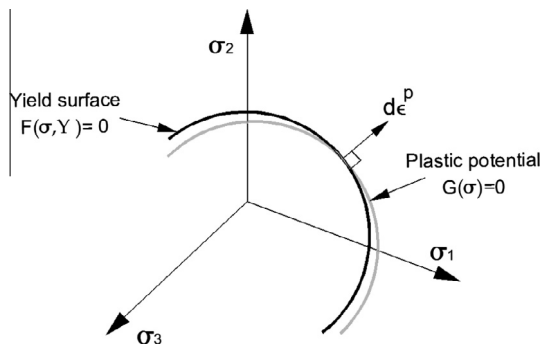


Fig. 4. Plastic potential surface and yield surface in the deviatoric plane.

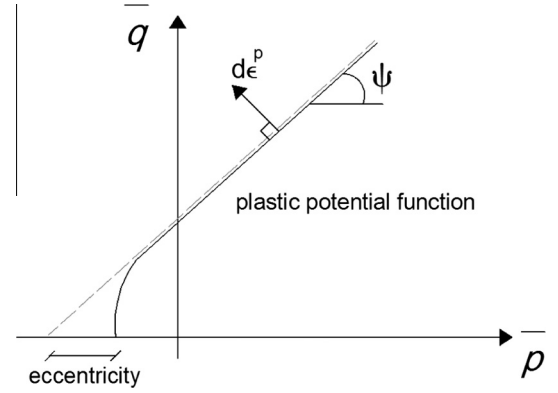


Fig. 5. Dilation angle and eccentricity in meridian plane.

The damage parameter d is defined in terms of compression and tension, d_c and d_t , respectively, such that:

$$(1 - d) = (1 - s_t d_t)(1 - s_c d_c) \quad (7)$$

where s_t and s_c describe the tensile and compressive stiffness recovery.

Viscoplastic regularization according to the Devaut–Lions approach can be introduced in the model. By defining the viscous parameter μ the plastic strain tensor is upgraded and the damage is deduced using additional relaxation time. Eq. (8) describes the strain rate with the viscoplastic regularization.

$$\dot{\varepsilon}_v^{pl} = \frac{1}{\mu} (\varepsilon_v^{pl} - \varepsilon_v^p) \quad (8)$$

Likewise, the viscoplastic damage increment is determined in Eq. (9):

$$\dot{d}_v = \frac{1}{\mu} (d - d_v) \quad (9)$$

where d_v denotes the viscous stiffness degradation variable. The relationship between stress and strain according to the viscoplastic model is given in Eq. (10).

$$\sigma = (1 - d_v)E_0 : (\varepsilon - \varepsilon_v^{pl}) \quad (10)$$

3.3. Material modeling

The concrete material parameters that were used in the presented analyses are: the modulus of elasticity E_0 , the Poisson's ratio ν and the compressive and tensile strengths of the selected slabs. The concrete damaged plasticity model considers a constant value for the Poisson's ratio, ν , even for cracked concrete. Therefore, in the analyses presented herein, the value $\nu = 0$ was assumed. The dilation angle ψ was considered as 40° , the shape factor, $K_c = 0.667$, the stress ratio $\sigma_{b0}/\sigma_{c0} = 1.16$ and the eccentricity $\varepsilon = 0.1$.

The uniaxial stress–strain response of concrete in tension is linear elastic up to its tensile strength, f'_t . After cracking, the descending branch is modeled by a softening process, which ends at a tensile strain ε_u , where zero residual tensile strength exists (Fig. 7). The concrete's brittle behavior is often characterized by a stress–crack displacement response instead of a stress–strain relationship. The stress–crack displacement relationship can be defined with different options: linear, bilinear or exponential tension softening response. In this study, bilinear stiffening response was used and was calculated according to the Fig. 6, where, f'_t is the maximum tensile strength and G_f denotes the fracture energy of

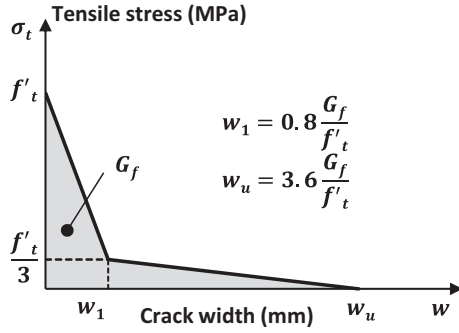


Fig. 6. Uniaxial tensile stress–crack width relationship for concrete.

concrete that represents the area under the tensile stress–crack displacement curve.

The fracture energy G_f depends on the concrete quality and aggregate size and can be obtained from Eq. (11) (CEB-FIP Model Code 90) [24].

$$G_f = G_{f0} (f_{cm} / f_{cm0})^{0.7} \quad (\text{N/mm}) \quad (11)$$

where $f_{cm0} = 10$ MPa and G_{f0} is the base fracture energy depending on the maximum aggregate size, d_{max} . The value of the base fracture energy G_{f0} is 0.026 N/mm for maximum aggregate size d_{max} equal to 10 mm that was used in the tested specimens.

According to [24], f_{cm} is the mean compressive strength of concrete and its relationship with the characteristic value, f_{ck} , is: $f_{cm} = f_{ck} + 8$ MPa.

In order to minimize the localization of the fracture, the tensile strains were used and they were defined by dividing the cracking displacement (w) by the characteristic length of the element (l_c).

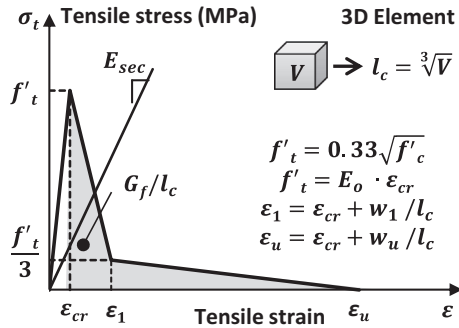


Fig. 7. Uniaxial tensile stress–strain relationship for concrete.

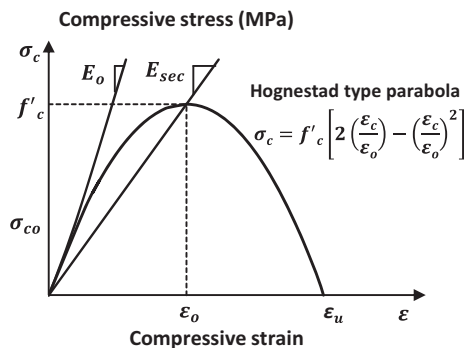


Fig. 8. Uniaxial compressive stress–strain relationship for concrete.

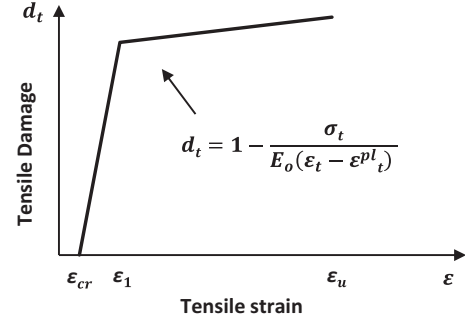


Fig. 9. Tensile damage parameter–strain relationship for concrete.

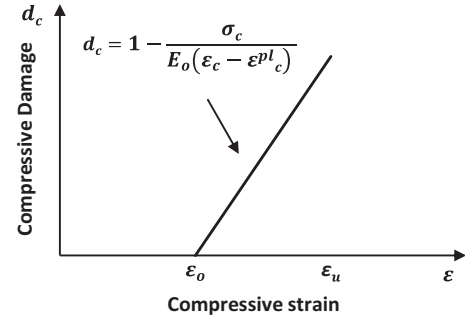


Fig. 10. Compressive damage parameter–strain relationship for concrete (simplified in linear form).

For 3D elements the characteristic length can be defined as the cubic root of the element's volume. The adopted critical length (l_c) in the following simulations was 20 mm. The tensile stress–strain graph is illustrated in Fig. 7.

Concrete in compression was modeled with the Hognestad parabola (Fig. 8). The assumed stress–strain relation behavior of the concrete under uniaxial compressive loading can be divided into three domains. The first one represents the linear-elastic branch, with the initial modulus of elasticity, $E_o = 5500 \sqrt{f'_c}$. The linear branch ends at the stress level of σ_{co} that here was taken as: $\sigma_{co} = 0.4 f'_c$. The second section describes the ascending branch of the uniaxial stress–strain relationship for compression loading to the peak load at the corresponding strain level, $\epsilon_o = 2 f'_c / E_{sec}$. The secant modulus of elasticity was defined as: $E_{sec} = 5000 \sqrt{f'_c}$. The third part of the stress–strain curve after the peak stress and until the ultimate strain ϵ_u represents the post-peak branch. The equation for the assumed compressive stress–strain diagram is given in Fig. 8.

Damage was introduced in concrete damaged plasticity model in tension and compression according to Figs. 9 and 10, respectively. Concrete damage was assumed to occur in the softening range in both tension and compression. In compression the damage was introduced after reaching the peak load corresponding to the strain level, ϵ_o .

The uniaxial stress–strain relation of reinforcement was modeled as elastic with Young's modulus (E_s) and Poisson's ratio (ν) of which typical values are 200,000 MPa and 0.3, respectively. Plastic behavior was defined in a tabular form, including yield stress and corresponding plastic strain. The plastic properties were defined based on the test results with a bilinear strain hardening yield stress – plastic strain curve. Table 2 presents the material properties of the reinforcement.

Table 2
Material properties of the reinforcement.

	Slab specimen	f_y (MPa)	ε_y	f_t (MPa)	ε_t
Interior	SB1	455	0.0023	650	0.25
	SW1, SW5	470	0.0024	650	0.20
Edge	XXX, HXXX (compression)	430	0.0022	600	0.15
	XXX, HXXX (tension)	545	0.0027	900	0.10

4. Investigation on material parameters and calibration of the model

At the outset, it is essential to discuss the chosen material parameters. Slab SB1 was chosen as a control slab in order to investigate the parameters of the concrete damaged plasticity model in ABAQUS and to calibrate the model. Two types of analyses were performed, static analysis in ABAQUS/Standard with the viscosity regularization and quasi-static analysis in ABAQUS/Explicit. The solution procedure that ABAQUS/Explicit uses is dynamic, but it can also be used for static solutions with low rate of loading [16]. This type of analysis is called quasi-static and is appropriate for nonlinear problems such as punching shear, where the cracking of concrete leads to stiffness reduction.

Fig. 11 presents the comparison between static and quasi-static analyses compared with the experimental results in terms of force–displacement response. Two values for the viscosity parameter for the static analysis were used; $\mu = 0.00001$ and $\mu = 0.00005$. The value of the viscosity parameter depends on the time increment step and according to [25], μ should be around to 15% of the time increment step in order the solution to be improved without changing the result. In the SB1 analyses, due to the high nonlinearity of the problem, the time increment step could not be fixed and thus this guideline could not be directly used to define the viscosity parameter. The time increment step was set as automatic and the viscosity parameter was found through the numerical investigation. The results from both analyses, static with viscosity parameter equal to 0.00001 and quasi-static, are in good agreement compared to the tested results in terms of ultimate load and deflection (see Fig. 11). Brittle punching shear failures were observed from the analyses with the sharp peak in the load–deflection diagrams. Quasi-static analysis requires less computational time compared to the static analysis with viscoplastic regularization and provides good results. In all subsequent analyses the quasi-static analysis was used for all specimens.

A mesh convergence study was performed in our model. The model appears strain localization that relies on the smeared crack approach. When the strain localization accumulates within few elements, the remainder of the construction starts to unload. Finer

mesh leads to narrow band of localization and after a while the equations fail to converge numerically. For that reason the model becomes mesh size dependent as happens with most plasticity based models that appear strain softening. Between the various ways to remedy the mesh size dependency due to the spurious strain-softening localization are: the introduction of the characteristic internal crack length at the softening part of the stress–strain relationship into the constitutive model and the viscoplastic regularization. Both approaches were used in our study. Three different mesh sizes (15 mm, 20 mm and 24 mm) were adopted in the analysis of the slab in order to investigate the mesh sensitivity of the model. These values were chosen in order for the meshing elements to be larger than the aggregate size (10 mm) and also not too large resulting in a coarse mesh. By having elements with mesh size of 15 mm, 8 elements were considered through the thickness of the slab, while by having mesh sizes of 20 mm and 24 mm, 6 and 5 elements were considered through the thickness of the slab, respectively. The choice of 24 mm mesh size (5 elements through the slab's thickness) was adopted in order to avoid the hourglassing numerical problem and the distortion that happen in C3D8R elements with coarse mesh. Even if tensile strains were used by dividing the cracking displacements to the characteristic lengths of the elements, the results remained mesh size dependent. In Fig. 12 the analyses are presented with 15 mm, 20 mm and 24 mm mesh sizes. The results are mesh size dependent, especially in terms of failure displacements. All the mesh sizes give similar results in terms of the failure load. The mesh size of 20 mm gave the most accurate results compared to the test data. The mesh size of 24 mm seems to be too coarse and not to converge giving a ductile and not realistic behavior to the slab. The mesh size of 15 mm seems to be too small (close the aggregate size) and for that reason it cannot be considered. These observations were made after performing quasi-static analysis. The results with the viscoplastic regularization in ABAQUS/Standard by performing static analysis for the same mesh sizes (15 mm, 20 mm and 24 mm) were still mesh size dependent. The viscous parameter was obtained as a material property for the concrete, introducing rate dependence into the material as relaxation time (Eq. (10)). The value of the chosen viscous parameter was 0.00001. The consideration of the viscoplastic component in the model did not seem to fully resolve the mesh sensitivity of the problem. However, the mesh size of 20 mm gave the most accurate results compared to the test load–deflection response.

The mesh size of 20 mm was chosen in all subsequent simulations of all slabs. This choice has been based not only on the load–deflection responses but also on the comparisons with the tested cracking patterns.

Concrete as a brittle material undergoes considerable volume change caused by inelastic strains. This volume change is called

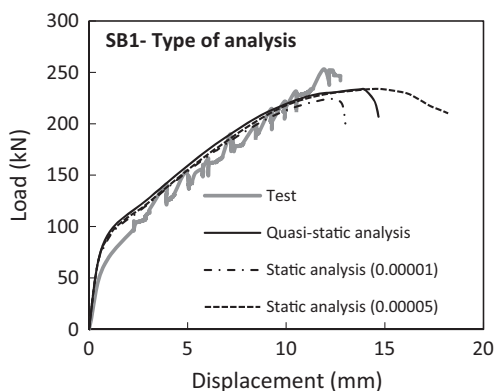


Fig. 11. Load–deflection response of slab SB1 by comparing static and quasi-static analyses.

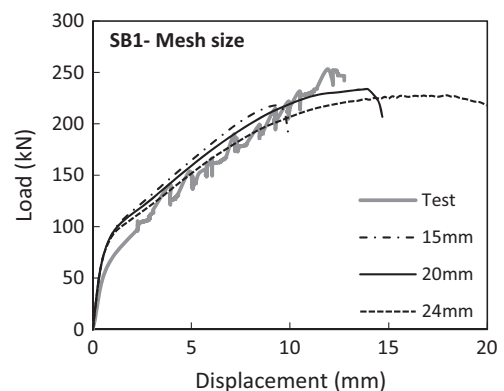


Fig. 12. Load–deflection response of slab SB1 for different mesh sizes (quasi-static analysis).

dilatancy. In concrete damaged plasticity model the dilatancy can be modeled by defining a value for the dilation angle. According to [7], the non-associated flow rule should control the dilatancy, especially for frictional materials such as concrete. Therefore, dilation angle can be considered as a material parameter for concrete. Lee and Fenves [23] defined the dilatancy parameter α_p equal to 0.2 in the Drucker–Prager plastic potential function (Eq. (12)).

$$G = \sqrt{2J_2} + \alpha_p I_1 \quad (12)$$

Concrete damaged plasticity model uses Eq. (5) for the flow potential function, which derives from Eq. (12). Therefore, Eq. (5) can be rewritten as:

$$G(\sigma) = \sqrt{(\varepsilon \sigma_{i0} \tan \psi)^2 + \bar{q}^2} + \frac{1}{3} I_1 \tan \psi \quad (13)$$

The above equations describing the plastic potential function result in the dilation angle (ψ) of 31° for $\alpha_p = 0.2$. The same dilatancy parameter was used by the same authors in Reference [25]. Other researchers (Wu et al. [26] and Voyiadis and Taqieddin [27]) defined the parameter α_p to range between 0.2 and 0.3. Therefore, the dilation angle (ψ) in concrete damaged plasticity model should range between 31° and 42° . Herein, the dilation angle for the model was examined with values varying from 20° to 42° (see Fig. 13). The dilation angle of 40° was chosen for the following analyses after investigation shown in Fig. 13. It can be shown that the difference in ultimate load is small between 38° and 42° . Therefore, the dilation angle was chosen to be set as 40° for all specimens.

Fig. 14 presents the results obtained by performing analyses with different values for the parameter K_c that gives the shape to the yield surface. According to [17] the parameter K_c should satisfy the condition: $0.5 < K_c \leq 1$ and the default value that is should be given for it; is 0.667. Three different values were given for investigation of the parameter, K_c : 0.667, 0.9 and 1. The results in Fig. 14 indicate that the difference in the load–displacement response of the slab–column connection by giving different shape to the yield surface is not significant. Considering K_c equal to 1, the simulation gives stiffer results and as the parameter K_c is getting less the load and the ultimate displacement are going to be increased. Consequently, for all the next analyses the parameter K_c was defined with its default value of 0.667.

The investigation on the influence of the damage parameters is shown in Fig. 15. The damage parameters in concrete damaged plasticity model take into consideration the degradation of concrete after cracking. The maximum value for the damage parameters in both tension and compression was chosen to be 0.9. The tensile damage parameter at the strain level ε_1 (Fig. 9) was chosen to be set as 0.85. For the definition of the compressive damage

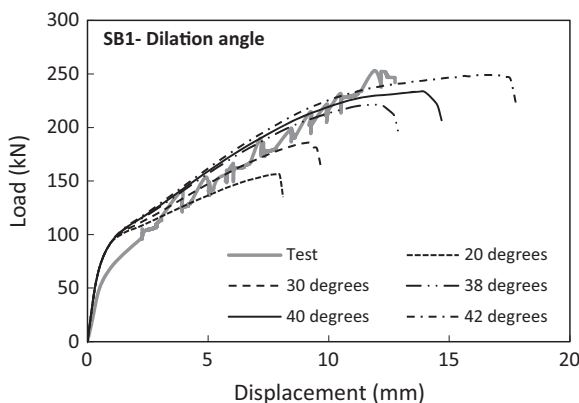


Fig. 13. Load–deflection response of slab SB1 for different values of dilation angle.

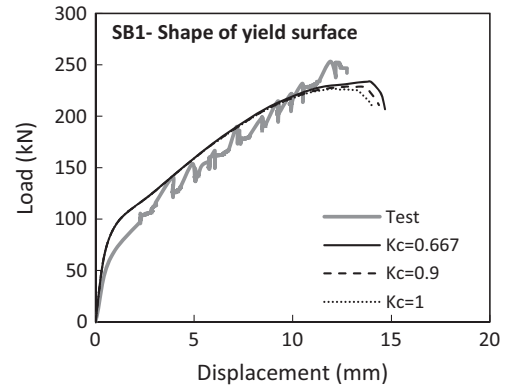


Fig. 14. Load–deflection response of slab SB1 for different shapes of the yield surface.

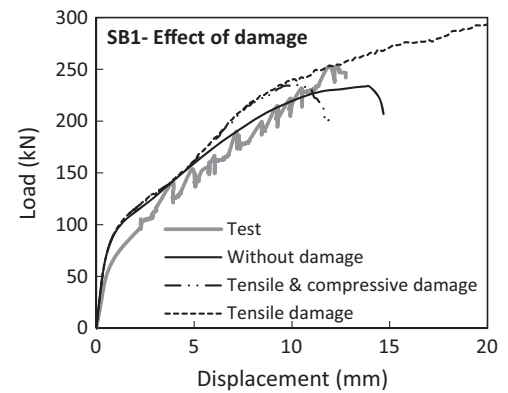


Fig. 15. Load–deflection response of slab SB1 (with and without damage).

parameters simplified linear relationship was adopted by given the minimum damage parameter equal to zero at the strain level ε_o and the maximum value 0.9 at the strain level ε_u (Fig. 10). The results obtained from the analysis considering the damage parameters displayed that the failure of the control specimen SB1 happened earlier compared to the analysis results without considering damage parameters. This becomes clear if one realizes that, the plastic strains are lower compared to the inelastic strains. The latter were considered in the model without the definition of the damage parameters. Without considering the damage parameters, the model behaves with only plasticity, assuming the plastic and inelastic strains to be equal. It could be noted that the damage had no effect at the early stage when the load was 100 kN (see Fig. 15). This happened because the concrete at this load was almost elastic and no or little damage had occurred. If only tensile damage is considered the results overestimate the ultimate loading capacity of the slab. When the damaged model was applied to both tension and compression, the model appeared to underestimate the ultimate load of the slab–column connection. Damage in compression seemed to have significant effect on the numerical results. Based on the observations of the effect of the damage parameters, it can be said that the damage parameters in the concrete damaged plasticity model in ABAQUS are similar to the hardening parameters used in the classic plasticity theory. For the described problem of punching shear the definition of the damage parameters should not be taken into consideration even if the numerical results underestimate the loading capacity. It is supposed that the damage parameters are important for cyclic or dynamic loadings where unloading should be defined by plastic strains.

The fracture energy of concrete can be related to the strength of concrete. For the 44 MPa strength of concrete for SB1 specimen, Eq.

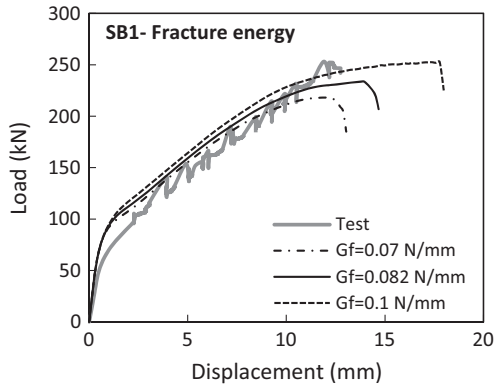


Fig. 16. Load–deflection response of slab SB1 with different values of fracture energy.

Table 3
Details of the simulated slabs in ABAQUS.

	Slab specimen	Type of loading	f'_c (MPa)	f'_t (MPa)	E_c (MPa)	G_f (N/mm)
Interior	SB1	Static	44	2.2	36,483	0.082
	SW1	Static and reversed cyclic	35	2	32,538	0.072
	SW5	Static and reversed cyclic	46	2.2	37,303	0.085
Edge	XXX	Static and horizontal	33	1.9	31,595	0.081
	HXXX	Static and horizontal	36.5	2	33,228	0.085

(11) gives the fracture energy equal to 0.082 N/mm. Fig. 16, illustrates the influence of the fracture energy on the slabs' response. Three different values (0.07 N/mm, 0.082 N/mm and 0.1 N/mm) were studied. The different responses depending on the value of the fracture energy show that the contribution of the tensile behavior of the concrete to the response of the slab is significant, which is logical since punching shear failure for slabs without shear reinforcement is dependent on the tensile response of concrete. For the following analyses, the fracture energy of 0.082 N/mm was used for the slab specimen SB1. Thus, Eq. (11) was used for defining the values of fracture energy for all other specimens.

Table 3 provides the summary of the concrete material parameters used in concrete damaged plasticity model in ABAQUS for each slab-column connection and details regarding the type of the connection and loading.

All tested specimens failed in punching shear. The information regarding their failure loads and comparisons with the simulation results are presented in the next sections.

5. Finite element analyses results

5.1. Control specimen SB1

Fig. 17 presents the final analysis results in terms of load–displacement for slab SB1. The simulation gives brittle punching shear failure as in the experiment. The ultimate load and displacement predicted by the simulation and the test are presented in Table 4. The FEA shows stiffer response than the test; possibly, due to the initial micro-cracking of the slab prior to testing. The cracking pattern on the tension side of the slab at failure is presented in Fig. 18. The cracking propagates inside the slab adjacent to the column. It starts tangentially near the column and then extends radially as the load increases. At the ultimate load the punching shear cone is visible due to the sudden opening of the cracks. Concrete

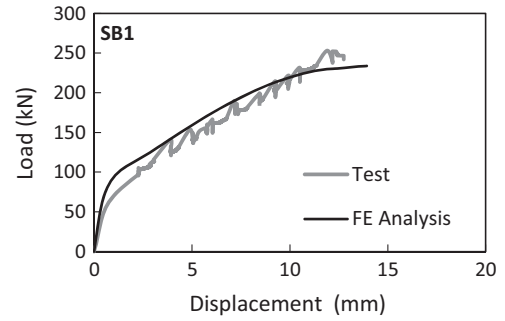


Fig. 17. Load–displacement curves for slab SB1.

Table 4
Test and FEA results.

Slab specimen	Test results		FEA results	
	Failure load (kN)	Displacement at failure (mm)	Failure load (kN)	Displacement at failure (mm)
SB1	253	11.9	234	13.9

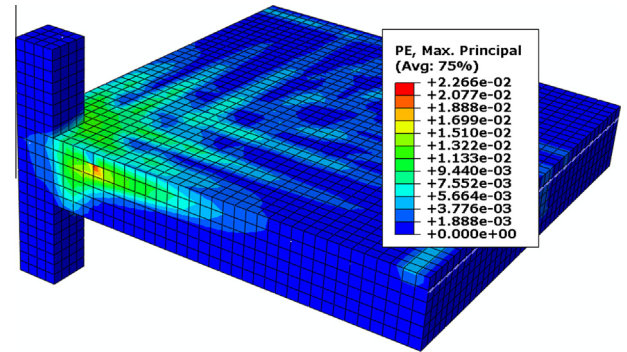


Fig. 18. Cracking pattern on tension surface at ultimate load for slab SB1 (load applied upwards on the column).

damaged plasticity model assumes that the cracking initiates when the maximum principal plastic strain is positive. The orientation of the cracks is considered to be perpendicular to the maximum principal plastic strains and thus, the direction of the cracking is visualized through the maximum principal plastic strains (Fig. 18). The yielding of the flexural reinforcement has occurred at the failure in both test and FEA. The tensile longitudinal reinforcement yielded under the column at the failure load.

The maximum tensile principal stresses are shown in Fig. 19 for the two surfaces of the slab at the failure. The tensile principal stresses can be used in FEA in order to show the cracking patterns. However, the maximum plastic equivalent principal strains as they were presented in Fig. 18, give a better representation of the cracks. For that reason the strains are going to be used for showing the cracking patterns for all the following analyses.

5.2. Slab-column connections SW1 and SW5

The calibrated model for specimen SB1 was then applied for the analyses of slabs SW1 and SW5. These slabs were tested under gravity load and horizontal reversed cyclic displacements. The response of the specimens is described by means of horizontal load and drift response. The test hysteretic loops in the specimen exhibited pinching, denoting strength and stiffness degradation and subsequently low energy dissipation capacity. In contrast, when the full dynamic analysis was performed in ABAQUS, the hysteretic

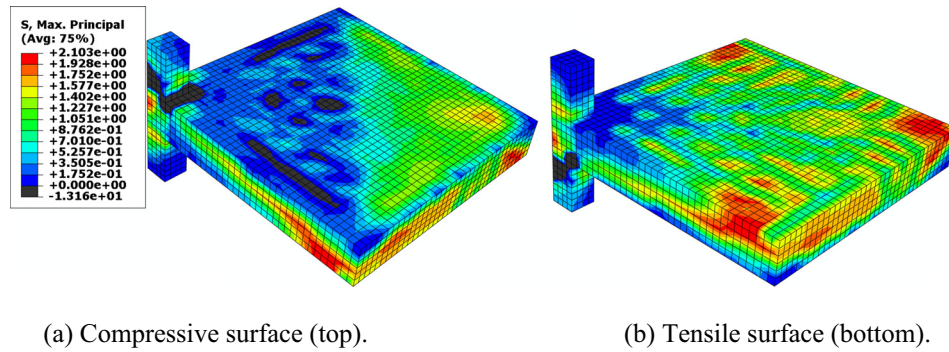


Fig. 19. Maximum tensile principal stresses in concrete at the failure.

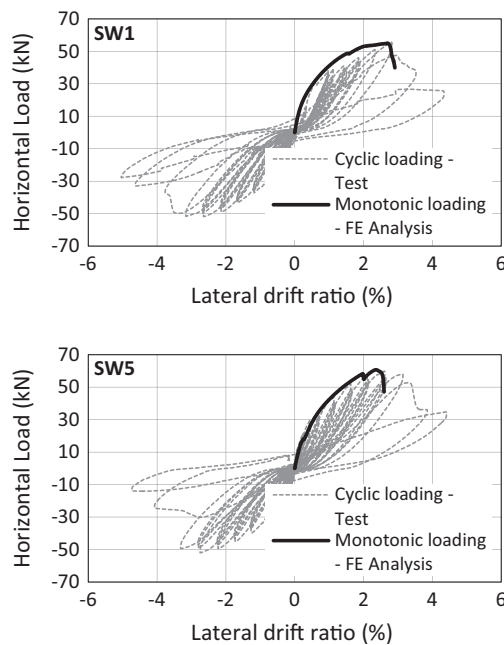


Fig. 20. Horizontal load–lateral drift ratio for specimens SW1 and SW5.

loops obtained from the analyses did not exhibit the pinching effect. It must be mentioned that the complexity in constitutive modeling of concrete and the adoption of perfect bond between concrete and reinforcement, created problems in the hysteretic simulations in ABAQUS. Alternatively, in this paper, monotonic loading analysis is presented and the results of the finite element simulations show good agreement compared to the experimental results (Fig. 20). Simulations of specimens show brittle failure after obtaining maximum lateral load similar to the test maximum loads. Fig. 21 presents the cracking pattern at failure for each specimen. Table 5 compares the experimental and numerical results in terms of ultimate lateral failure load and drift ratio at the failure load. Yielding of the flexural reinforcement during the test, for the specimen SW1 appeared first at the tension reinforcement under the column in the direction of the cyclic loading at drift ratio 1.33%, while for the specimen SW5 at the compression reinforcement under the column in the direction of the cyclic loading at drift ratio 1.04%. The FEA results have proved similar yielding of the flexural reinforcement.

5.3. Slab-column connections XXX and HXXX

The edge slab-column specimens were examined using the FEA model identical to the one used for SB1. These slabs, tested under

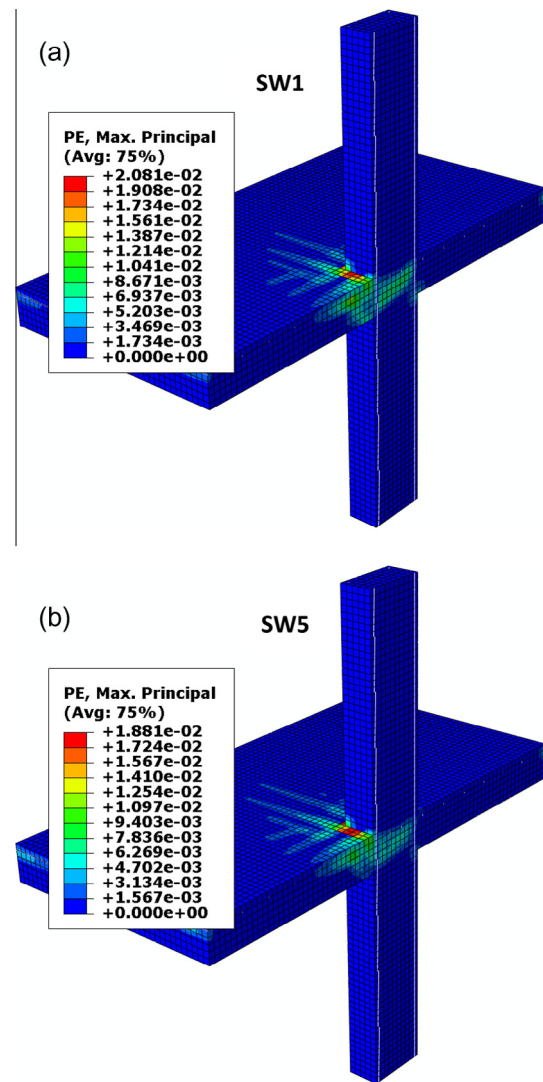


Fig. 21. Cracking pattern at ultimate load for specimens: a) SW1 and b) SW5.

constant gravity load to horizontal moment ratios, providing information for the effect of the unbalanced moments on punching shear. Table 6 shows the comparison between the slabs XXX and HXXX in terms of failure horizontal load and displacement and subsequently compares the tested and FEA results. Fig. 22 presents the load–displacement analytical results compared to the test results. The simulated response of the specimen XXX, in terms of ultimate load and displacement, is in good agreement with the

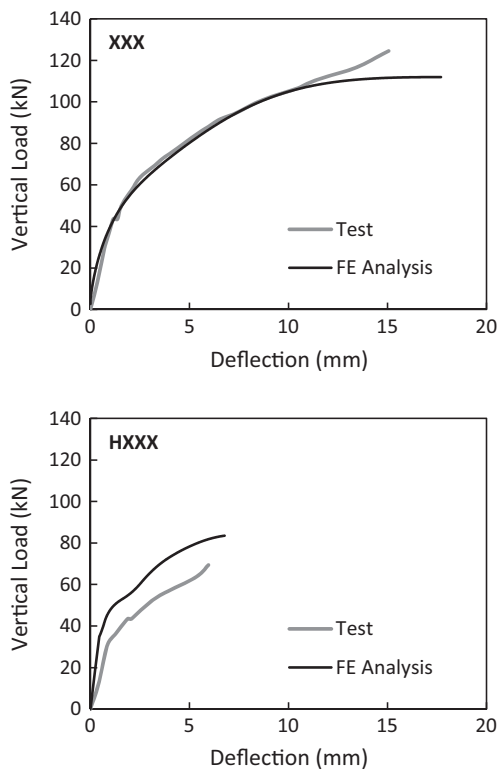
Table 5

Gravity shear ratio, test and FEA results.

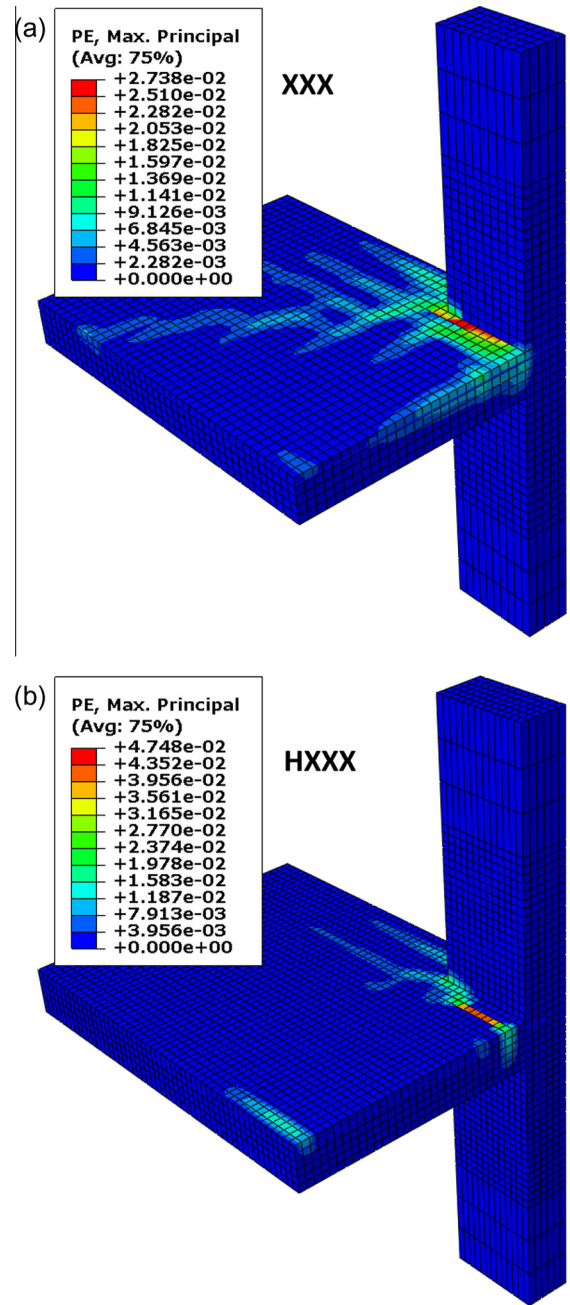
Slab specimen	V/V_n	Test results		FEA results	
		Peak lateral load (kN)	Drift ratio at peak lateral load (%)	Peak lateral load (kN)	Drift ratio at peak lateral load (%)
SW1	0.54	56	2.8	55	2.7
SW5	0.68	60	2.6	61	2.4

Table 6 M/V ratios, test and FEA results.

Slab specimen	M/V (m)	Test results		FEA results	
		Failure horizontal load (kN)	Displacement at failure load (mm)	Failure horizontal load (kN)	Displacement at failure load (mm)
XXX	0.3	125	15.06	112	17.69
HXXX	0.66	69	5.96	84	6.77

**Fig. 22.** Vertical load–deflection for edge slabs; XXX and HXXX.

results observed from the experiment (within 10% error). However, the relative error is within 20% between FEA and test results for the specimen HXXX. This is an acceptable difference and can be attributed to many reasons. One reason is that as it is shown in Fig. 22, the numerical load–deflection response of specimen HXXX appears stiffer compared to the tested response due to the possible initial pre-cracking prior to the test (e.g. shrinkage, handling). It is important to point out that the FEA results of slab HXXX, in terms of the failure displacement, were in good agreement with the tested results. The obtained FEA cracks of HXXX specimen were similar to the tested cracks, concentrated near the column with some developed radial cracks. The cracking propagation at the ultimate load for both slabs is presented in Fig. 23. Comparison between

**Fig. 23.** Cracking pattern at the ultimate on the tension surface for edge slabs: a) XXX and b) HXXX.

the predicted crack patterns of slabs XXX and HXXX; shows the effect of the higher moment at the slab–column connection. The increased unbalanced shear load and deflection and thus resulted to a more sudden and brittle punching shear failure. The tensile longitudinal reinforcement under the column has yielded in tests and FEA for both XXX and HXXX slabs. The reinforcement of the slab XXX yielded at a load of 78 kN and the reinforcement of the slab HXXX yielded at a load of 48 kN during the tests. The FEA showed almost the same results. The reinforcement of the slab XXX yielded at a load of 73 kN and the reinforcement of the slab HXXX yielded at a load of 55 kN. Good agreement was observed for the activation of the flexural reinforcement in the test and analysis before and after the yielding.

6. Discussion and conclusions

In this paper, the finite element analysis with the concrete damaged plasticity model was used for predicting punching shear response of slabs without shear reinforcement. In particular, five different slab-column connections without shear reinforcement were simulated and analyzed in terms of ultimate load and cracking patterns. The constitutive formulation adopted herein, is a damaged plasticity model implemented in the finite element code, ABAQUS. This constitutive model has been first calibrated using experimental results of one of the slabs, SB1; an interior slab-column connection that was tested under concentric punching. The calibrated model was then, used for the analyses of interior slab-column connections subjected to gravity and lateral displacements, and for the analyses of slab-column edge connections, all of which were previously tested at the University of Waterloo [19–21]. The results of the analyses compared to the test results, showed good agreement.

The most challenging aspect in finite element modeling of concrete structures is the accurate material modeling and especially the modeling of concrete. The parametric investigation was performed in both ABAQUS/Standard and ABAQUS/Explicit in order to calibrate the material model given in ABAQUS. Many material parameters were studied, among them, the dilation angle and the use of the damage parameters appear to be critical for the accurate definition of the concrete modeling. Likewise, the mesh sensitivity analysis is essential for providing the most appropriate element size due to mesh size-dependent model. The mesh size dependent issue was addressed and possible remedies using the characteristic length and viscoplastic regularization, as it was discussed. It was observed that the cracking propagation together with the load–displacement response should be taken into consideration for adoption of proper mesh size. Taking into consideration the parametric investigation for the material modeling, the analysis results give accurate punching shear prediction.

The finite element analysis results confirm the ability of the proposed model for predicting the punching shear failure in concrete slabs without shear reinforcement. The importance of finite element analysis as an assessment tool is that it can provide insight into punching shear failure and crack formation and allows for parametric studies, which cannot be obtained through experimental investigations. The presented analyses indicate that the proposed model can be used in future parametric studies on different aspects influencing punching shear in concrete slabs.

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